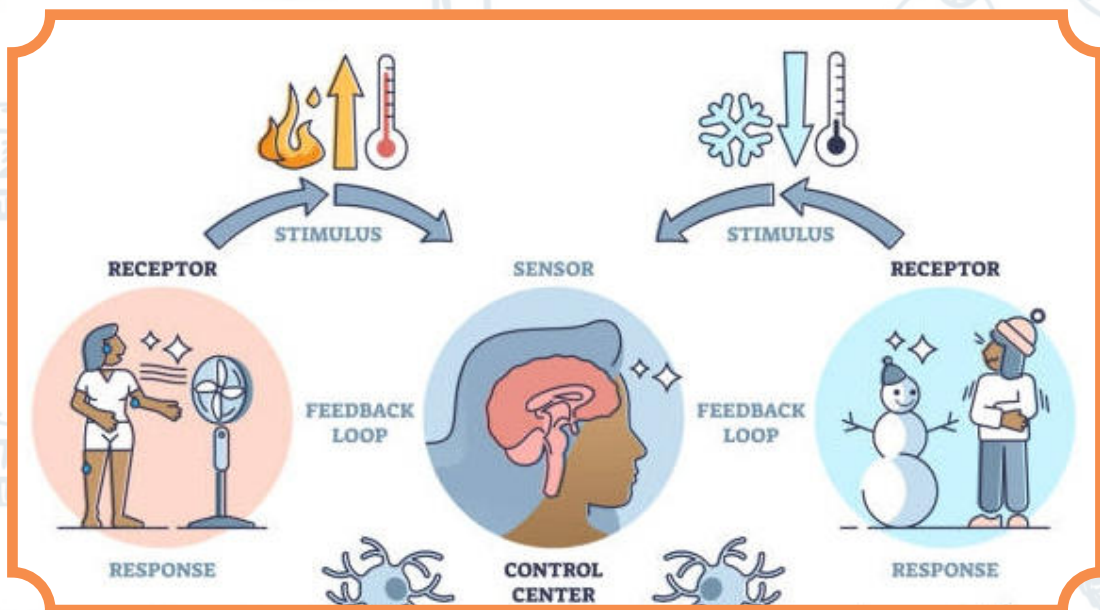


UNIT

15

HOMEOSTASIS





Definition:

The term **homeostasis** was introduced by **Walter B. Cannon** (1871–1945). It refers to the **self-regulating process** by which biological systems maintain stability while adjusting to changing environmental conditions. Homeostasis is essential for the **continuity of life**, as it helps to maintain the stability of body functions, keeping them within suitable limits despite external changes.

Importance of Homeostasis:

- **Stability of Body Functions:** It maintains the internal environment of the body, such as temperature, pH, and ion concentrations, within narrow limits that are optimal for cell function.
- **Adaptation to Environmental Conditions:** Organisms need to adapt to both moderate and extreme environmental conditions, and homeostasis ensures that internal conditions remain stable, even when external conditions fluctuate.

Components of Homeostasis:

Homeostasis involves maintaining a stable **internal environment**, which includes:

- **Body Fluids:** Blood, lymph, interstitial fluid.
- **Tissues, Organs, and Systems:** Various physiological systems work together to maintain homeostasis.

Regulation of Homeostasis:

The processes that regulate homeostasis in the body include:

- **Endocrine System:** Releases hormones to regulate functions such as metabolism, growth, and electrolyte balance.
- **Nervous System:** Controls and coordinates body functions through nerve impulses.

Homeostatic Mechanisms:

Several physiological processes in the body are controlled to maintain homeostasis, including:

1. Osmoregulation:

- Maintains the balance of water and salts in the body.
- Ensures that the concentration of solutes in body fluids remains within an optimal range.

2. Excretion:

- Removes waste products and excess substances from the body, primarily through the kidneys and the urinary system.
- Prevents toxic build-up that could disrupt body function.

3. Thermoregulation:

- Maintains the internal body temperature within a range that allows enzymes and other biochemical processes to function effectively.
- Mechanisms include sweating, shivering, and changes in blood flow.

Osmoregulation

Definition: Osmoregulation is the process by which organisms regulate the **balance of water and solutes** (such as salts and minerals) in their body to maintain **homeostasis**. It ensures that vital body functions, such as cellular processes, are not disrupted by fluctuations in external water or solute levels.





Types of Osmoregulation:

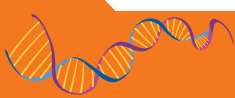
1. Osmoconformers:

- These organisms adjust their internal solute composition to match the external environment.
- They do not necessarily have the same solutes as the surrounding environment, but they maintain osmotic balance by accumulating certain solutes like **urea** and **trimethylamine oxide (TMAO)**.
- **Examples:** Marine organisms such as **sharks, marine invertebrates** (echinoderms like starfish, mussels, marine crabs, lobsters), **jellyfish**, and **ascidians**.
- **Key characteristic:** Osmoconformers are **isotonic** to their surroundings and can tolerate a wide range of salinities.



2. Osmoregulators:

- These organisms maintain a constant internal osmotic environment, **regardless** of external conditions.
- They actively regulate water and solute balance, even in fluctuating external environments (such as marine, freshwater, or terrestrial).
- **Examples:** Freshwater and terrestrial organisms, including **vertebrates** like fish and amphibians, and **land animals**.
- **Key characteristic:** Osmoregulators are adapted to function within a **narrow range of salinities**.





Problems Faced by Osmoregulators:

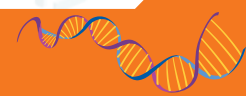
1. Freshwater Animals:

- Freshwater environments are **hypotonic** (lower solute concentration than the internal environment of the organism).
- Water tends to enter their body through osmosis, especially via the **skin** and **gills**, which can disturb their osmotic balance.
- Freshwater animals have adaptations to **excrete excess water** while **retaining salts** to maintain osmotic equilibrium.



2. Marine Animals:

- Marine environments are **hypertonic** (higher solute concentration than the internal environment).
- Marine animals risk **dehydration** due to water loss, so they must **drink seawater** to maintain hydration.
- The challenge is **retaining salts** from the seawater, which must be excreted to avoid disruption of the osmotic balance.
- Marine organisms have special **salt-excreting glands** to get rid of excess salts.





3. Terrestrial Organisms:

- Terrestrial environments lead to **water loss** due to **perspiration**, which can cause dehydration and loss of essential salts.
- These organisms must **retain water** while managing the loss of salts through **sweating** or **exhalation**.
- Adaptations like **kidneys** for water retention and **specialized salt glands** to excrete excess salts help to manage the osmotic balance.



Osmoregulation in Different Habitats:

1. Osmoregulation in Freshwater Habitat

Environmental Conditions:

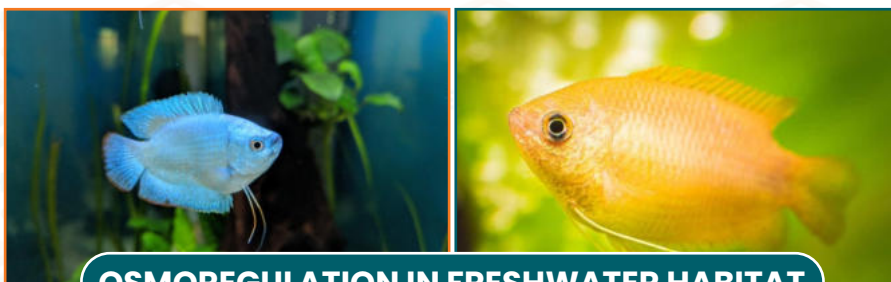
Freshwater animals face an external hypotonic environment (lower solute concentration) compared to their internal environment, which is hypertonic (higher solute concentration). This creates the problem of water influx and the need to retain salts.

Challenges:

- **Water influx:** Water tends to enter the body of freshwater animals by **osmosis** through their **skin** and **gills**.
- **Salt loss:** The external environment has a lower concentration of salts, causing animals to lose salts across their gills and skin.

Adaptations:

- **Kidneys:** Freshwater animals have **high filtration rates in their kidneys**, which produce **dilute urine** to excrete excess water while **retaining salts**. The kidneys are specialized to **actively reabsorb salts** from the filtrate, returning them to the bloodstream.
- **Ionocytes:** Specialized cells in the **gills** and **skin** called **ionocytes** help to **actively extract ions** (such as Na^+ , Cl^- , and Ca^{2+}) from the external medium and reintroduce them into the body. These cells also help in maintaining **acid-base balance** by excreting excess H^+ or HCO_3^- .





2. Osmoregulation in Marine Habitat

Environmental Conditions:

Marine animals live in a **hypertonic external environment** (higher solute concentration) compared to their **internal environment**, which poses the problem of **dehydration**.

Challenges:

- **Dehydration:** Marine animals lose water to the external environment due to the high salinity, which can lead to dehydration.
- **Salt gain:** Seawater also contains excess **salts**, which need to be managed.

Adaptations:

- **Drinking seawater:** Marine animals actively drink seawater to compensate for water loss.
- **Salt excretion:** They have specialized **salt-excreting glands** (like the **rectal glands** in sharks and **salt glands in seabirds**) to **actively excrete excess salts** without losing too much water.
- **Osmotic balance:** Marine organisms have adapted to regulate water intake and salt excretion efficiently, ensuring that they do not become overly dehydrated while keeping their internal solute concentration within limits.



Osmoregulation in Marine Water Habitat:

Environmental Conditions:

Marine animals live in a **hypertonic environment** (higher solute concentration) compared to their **internal body fluids**, meaning they tend to lose water and gain salts from the surrounding seawater.

Challenges:

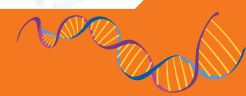
- **Water loss:** Marine animals lose water to the environment due to the osmotic difference.
- **Salt gain:** Marine animals need to manage the excess salts present in the seawater.

Adaptations:

- **Drinking seawater:** Marine animals, especially marine fishes, drink seawater to compensate for water loss.
- **Salt digestion:** The water they ingest also contains **marine salts**, which are **harder** and more concentrated than freshwater salts. Marine animals can **digest these salts** in a controlled way.

Key Osmoregulatory Organs:

- **Gills:** Marine bony fish use their **gills** to excrete excess salts, particularly **Na⁺, Cl⁻, Ca²⁺, and Mg²⁺**.
- **Kidneys:** The **kidneys** help by reabsorbing water and some salts, while excreting a small volume of **concentrated urine** with excess salts (mainly divalent ions).
- **Rectal glands:** Specialized rectal glands in the intestines help in the removal of **excess salts** absorbed with seawater.





- **Digestive tract:** Marine fish's digestive systems help process and balance the intake of seawater and its salts.

Process Overview:

Marine fish maintain their **body fluid balance** by absorbing seawater, processing and digesting its salts, and then **excreting excess salts** through specialized **rectal glands** and **salt glands** located in the **gills**. This system ensures that their internal fluid composition remains stable.

Comparison: Osmoregulation in Marine and Freshwater Fishes

Marine Fish:

- Drink seawater to replace lost water.
- Actively excrete excess salts through the gills and rectal glands.
- Produce small quantities of **concentrated urine** containing mostly salts (e.g., Na^+ , Cl^- , Mg^{2+} , Ca^{2+}).

Freshwater Fish:

- Absorb water through osmosis.
- Actively reabsorb salts in the **gills** and produce **dilute urine** to excrete excess water while retaining salts.



OSMOREGULATION IN MARINE AND FRESHWATER FISHES

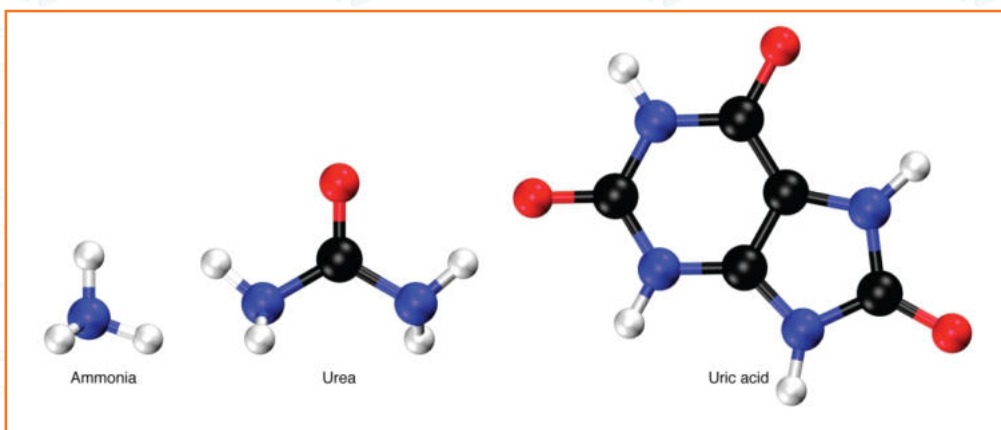
Excretion and Nitrogenous Wastes:

Excretion is the process of removing metabolic wastes generated from biochemical reactions in the body. It plays a vital role in homeostasis by controlling osmotic pressure, balancing inorganic ions, regulating water levels, and maintaining acid-base balance.

Nitrogenous wastes are produced from the breakdown of proteins and nucleic acids.

The three major types of nitrogenous waste are:

- Ammonia
- Urea
- Uric acid





Excretory Products in Relation to Habitat:

- **Ammonotelic:** Animals that excrete ammonia, primarily aquatic animals. Ammonia is highly toxic but soluble in water, which helps to dilute it in aquatic environments.
- **Ureotelic:** Terrestrial animals that excrete urea, which is less toxic and requires less water to excrete than ammonia.
- **Uricotelic:** Animals that excrete uric acid, such as reptiles, birds, and some arthropods. Uric acid is less toxic, requires even less water to excrete, and is beneficial in conserving water in arid environments.

Types of Nitrogenous Wastes:

1. Ammonia

- **Toxicity:** Highly toxic and corrosive, about 100,000 times more toxic than urea.
- **Excretion:** Ammonia is produced by the breakdown of nitrogenous compounds and is excreted in aquatic animals where it diffuses into the surrounding water. It requires large amounts of water to dilute and is primarily excreted in its ammonium (NH_4^+) form.
- **In terrestrial Animals:** Ammonia is converted to urea to reduce toxicity.



2. Urea

- **Production:** Formed in the liver through the urea cycle (ornithine cycle) by combining ammonia with carbon dioxide.
- **Excretion:** Excreted in urine by mammals, amphibians, some fish, and other ureotelic organisms.
- **Solubility:** Highly soluble in water.
- **Toxicity:** Less toxic than ammonia.
- **Function:** Urea plays a role in kidney function, particularly in the absorption of ions and water.
- **Benefits:** Used in the chemical industry to produce fertilizers.
- **Clinical Significance:** Blood nitrogen test (BUN) in humans measures the efficiency of renal function in eliminating urea and other nitrogenous wastes.





3. Uric Acid

- **Production:** An oxidative metabolic product of purines found in nucleic acids.
- **Solubility:** Less soluble in water compared to ammonia and urea.
- **Toxicity:** About 10,000 times less toxic than ammonia.
- **Excretion:** Excreted by uricotelic animals like reptiles, birds, and many arthropods. It is stored in tissues without harm and excreted as a thick paste.
- **Water Conservation:** Uric acid excretion allows animals to conserve water, as it requires little water to be excreted.
- **Humans:** In humans, 75% of uric acid is excreted by the kidneys, and 25% by the intestines.



Urinary System of Man:

The **urinary system** (also known as the **renal system**) is responsible for the removal of waste products and toxins from the body. It plays a crucial role in maintaining **homeostasis** by regulating:

- **Acid-base balance**
- **Electrolyte levels**
- **Blood pressure**
- **Blood pH**

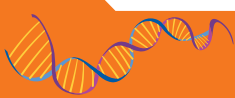
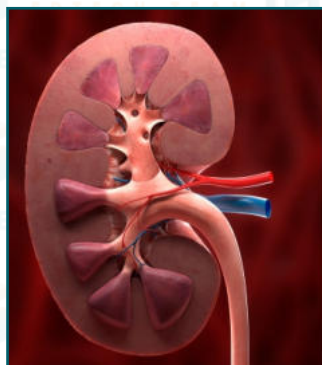
Main Organs of the Urinary System:

1. Kidneys

- **Function:** Filter blood to remove waste and excess substances, producing urine.
- **Structure:** Bean-shaped organs located in the abdominal cavity, near the adrenal glands.
- **Blood Supply:** Blood enters the kidneys via the renal arteries, and waste is filtered from the blood to form filtrate.
- **Filtrate:** The filtrate eventually becomes urine, which moves through the ureters.

2. Ureters

- **Function:** Transport urine from the kidneys to the urinary bladder.
- **Structure:** 25–30 cm long tubes that connect the kidneys to the bladder.
- **Structure:** Wide at the kidney end (renal pelvis) and narrow at the bladder end.



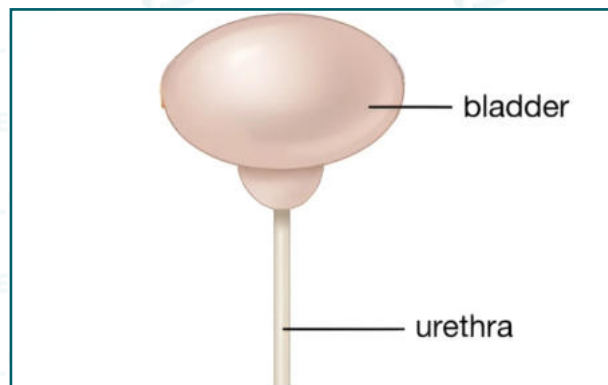


3. Urinary Bladder

- **Function:** Temporarily stores urine before it is excreted.
- **Structure:** Muscular and strong enough to hold about half a liter of urine in adults.
- **Control:** The bladder can store urine until it is full and ready to be excreted.

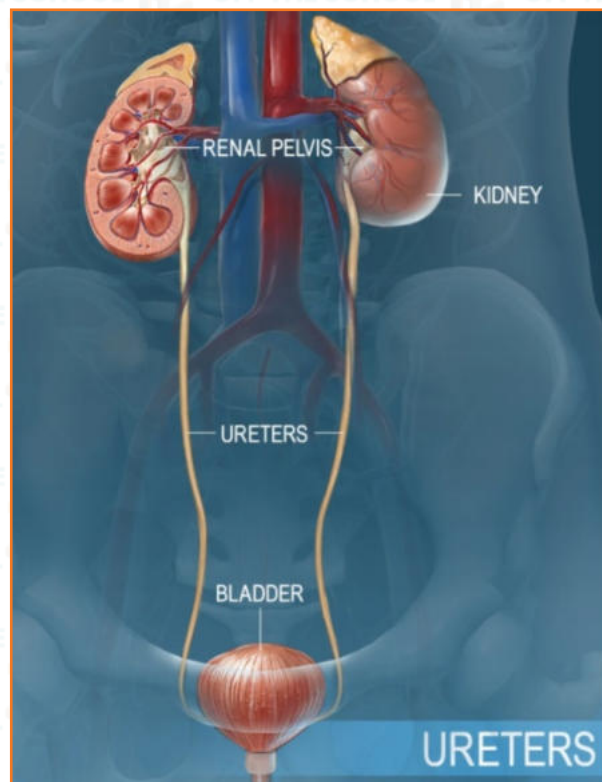
4. Urethra

- **Function:** A tube through which urine is expelled from the body.
- **Structure:** Thin, muscular, and controls the exit of urine.
Valves (Sphincters):
 - **Internal urethral sphincter:** Controls involuntary flow of urine from the bladder to the urethra.
 - **External urethral sphincter:** Provides voluntary control over the release of urine.



Blood Ammonia Levels

- Normal blood ammonia range: 15–45 $\mu\text{mol/L}$
- Levels above 100 $\mu\text{mol/L}$ can cause disturbance in consciousness. 200 $\mu\text{mol/L}$ is associated with coma and convulsions.





Structure and Function of Kidney

Location and Appearance:

- Kidneys are **symmetrical, bean-shaped**, reddish-brown organs.
- Located **below the rib cage**, on either side of the vertebral column between **T12 and L3 vertebrae**.
- Positioned **retroperitoneally** in the abdominal cavity.
- Approximately **4 to 5 inches long**, about the **size of a fist**.
- **Right kidney** is slightly **lower** than the left to accommodate the liver.
- **Left kidney** is slightly **larger** and **closer to the heart**.

External Structure:

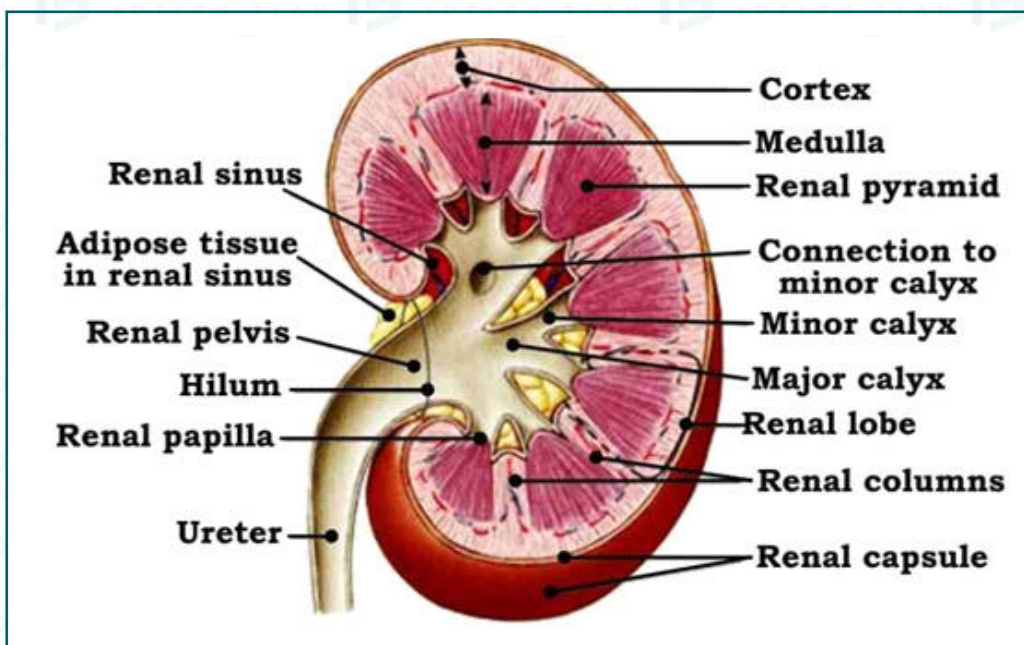
- Covered by **three protective layers**:
- 1. **Renal fascia** (outer tough connective tissue)
- 2. **Adipose (fatty) layer**
- 3. **Renal capsule** (inner protective covering)
- **Lateral side** is convex; **medial side** is concave forming the **renal sinus**.
- **Hilum**: the central concave region on the medial side through which **blood vessels, nerves, lymphatics, and the ureter** enter/exit.

Internal Structure:

- Divided into **two main regions**:
- 1. **Renal Cortex** (outer region):
 - Contains **arterioles, venules, and nephrons**.
 - Site of **erythropoietin production**—a hormone that stimulates red blood cell synthesis.
- 2. **Renal Medulla** (inner region):
 - Contains **7 to 18 renal pyramids** (conical-shaped tissue masses).
 - **Renal columns**: spaces between pyramids.
 - Mainly contains the **loop of Henle and collecting ducts** of nephrons.

Urine Formation and Flow:

- Urine forms in **nephrons**, flows through **collecting ducts** → moves to **renal pelvis** via pyramids → drained into the **ureter** for excretion.





Structure of Nephron

Basic Overview:

- Each kidney contains about **1 million nephrons**, the **functional units** responsible for **blood filtration** and **urine formation**.

- Nephrons are made up of two main parts:

1. **Renal Corpuscle** (site of filtration).
2. **Renal Tubule** (site of reabsorption and secretion).

Types of Nephrons:

1. **Cortical Nephrons.**

- Short **loop of Henle**, limited to the **renal cortex**.

2. **Juxtamedullary Nephrons.**

- Long **loop of Henle**, extends deep into the **renal medulla**.

- Important in **concentrating urine**.

Parts of a Nephron:

1. Renal Corpuscle (Malpighian Body):

- Composed of:

- **Glomerulus:** a ball of **capillaries** originating from the **afferent arteriole**.

- **Bowman's Capsule:** a thin-walled **cup-like structure** that encloses the glomerulus

- **Function:** Ultra-filtration.

- Glomerular capillaries have **pores (~70 nm)** to filter out water and small solutes.

- Large molecules and blood cells are **retained**.

- **Podocytes:** specialized cells with slit-like openings that prevent plasma proteins from passing into the filtrate.

- **Efferent arteriole:** carries blood **away** from the glomerulus.

2. Renal Tubule:

- **Proximal Convoluted Tubule (PCT):**

- Highly coiled, located in the **cortex**.

- Site for **reabsorption** of glucose, amino acids, salts, and water.

3. Loop of Henle:

- **Descending limb:** moves into medulla.

- **Ascending limb:** returns to cortex.

- Important in **concentrating urine**.

4. Distal Convoluted Tubule (DCT):

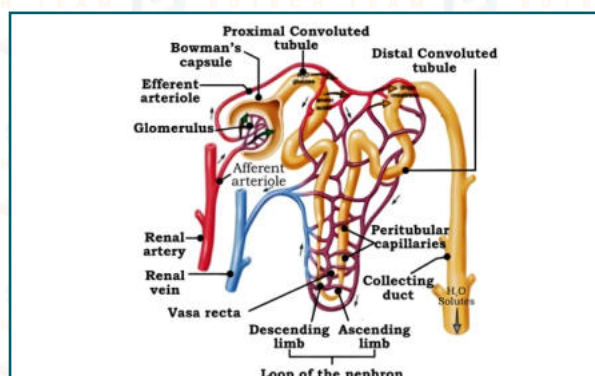
- Another coiled segment in the cortex.

- Involved in **secretion** and **fine-tuning** salt and water balance.

5. Collecting Duct:

- Receives urine from multiple nephrons.

- Passes through medulla and delivers urine to the **renal pelvis**.





Blood Supply to the Nephron

1. Entry of Blood:

- **Renal artery** enters the kidney at the **hilum**.
- It branches into **smaller arterioles**, including the **efferent arteriole** which supplies blood to the **glomerulus**.

2. Glomerular Capillary Bed:

- **Afferent arteriole** → forms dense capillary network called the **glomerulus**.
- Blood exits glomerulus via the **efferent arteriole**.
- **Afferent arteriole** is **wider** than the **efferent arteriole**, creating **high pressure** in glomerular capillaries for **efficient filtration**.

3. Peritubular Capillaries:

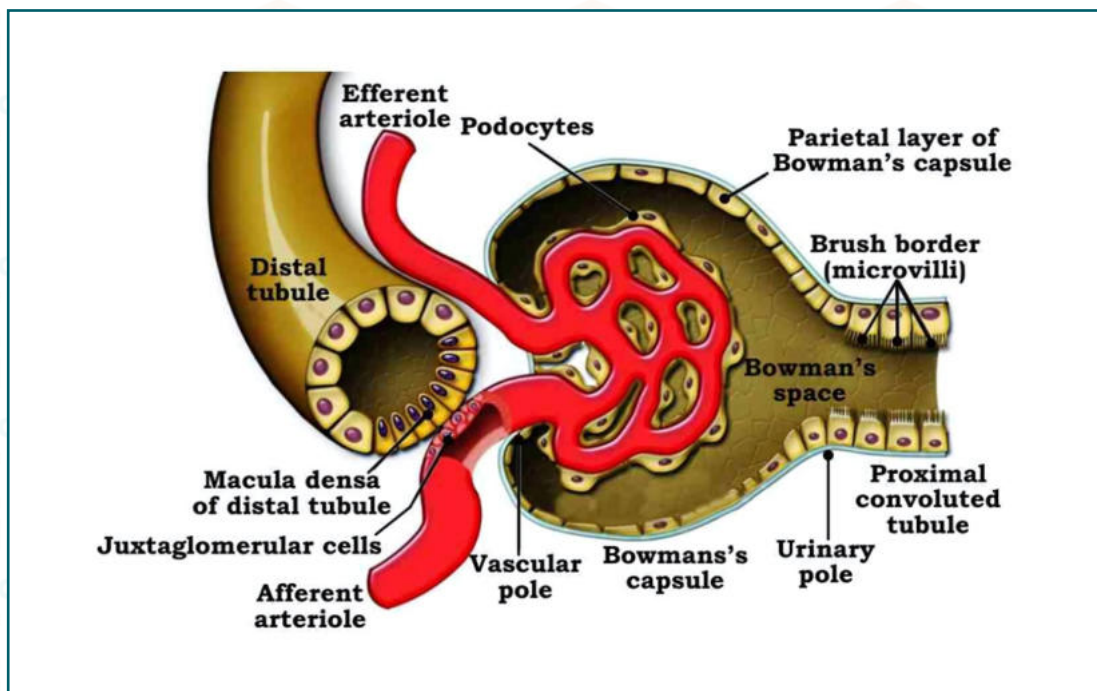
- **Efferent arteriole** extends and forms **peritubular capillaries** around:
- **Proximal Convoluted Tubule (PCT)**.
- **Distal Convoluted Tubule (DCT)**.
- These capillaries supply **oxygen and nutrients** to the **renal cortex**.

4. Vasa Recta:

- In juxtamedullary nephrons, efferent arterioles descend into the medulla to form **vasa recta**.
- **Vasa recta**:
- Run parallel to the **loop of Henle**.
- Involved in **countercurrent exchange** to maintain osmotic gradient in medulla.
- Connect with capillaries of the **renal venules**.

5. Venous Return:

- Blood from **peritubular capillaries** and **vasa recta** drains into **renal venules**.
- These venules merge into the **renal vein**, which exits the kidney and returns blood to systemic circulation.





Kidney Functions

- **Homeostatic organ:** Maintains internal balance (homeostasis).
- **Filtration:** Removes wastes and excess fluids via urine.
- **Endocrine function:**
- **Renin:** Regulates **blood pressure**.
- **Erythropoietin (EPO):** Stimulates **RBC production** in bone marrow.
- **Active Vitamin D₃ (Calcitriol):** Helps absorb **calcium** and **phosphorus** in intestines essential for **strong bones**.
- **pH regulation:** Maintains acid-base balance by:
 - Reabsorbing **HCO₃⁻ (bicarbonate)**.
 - Secreting **H⁺ (hydrogen ions)**.
- **Water balance:** Produces **dilute** or **concentrated urine** depending on hydration status and environment.

1. Plasma vs. Urine Composition

Substance	% in Plasma	% in Urine
Water	90%	95%
Protein	8%	0%
Glucose	0.1%	0%
Urea	0.03%	2%
Uric Acid	0.004%	0.05%
Ammonia	0.0001%	0.04%
Creatinine	0.001%	0.075%
Na ⁺	0.32%	0.35%
K ⁺	0.02%	0.15%
Cl ⁻	0.37%	0.60%
PO ₄ ³⁻	0.009%	0.27%
So ₄ ²⁻	0.002%	0.18%

2. Urine Formation – Three Main Steps

a) Glomerular Filtration:

- Occurs in the **glomerulus** under **high hydrostatic pressure** (due to afferent > efferent arteriole diameter).
- Produces **glomerular filtrate**: water, electrolytes, amino acids, glucose, bicarbonate, and nitrogenous wastes (urea, uric acid, creatinine).
- **GFR (Glomerular Filtration Rate):** ~180 L/day produces ~1.5–2.5 L of urine.
- **Autoregulated**, but may vary with fluid intake.





b) Selective Reabsorption:

- Occurs mostly in **Proximal Convoluted Tubule (PCT)**.

1. Reabsorbs:

- 100% Glucose**, amino acids, vitamins.
- ~67% Water, Na⁺**, variable **Cl⁻, Ca²⁺, Mg²⁺**
- Passive** reabsorption: Water.
- Active** reabsorption: Na⁺ and glucose.

2. Hormones:

- Aldosterone**: Enhances Na⁺ reabsorption.
- ADH (Vasopressin)**: Enhances water reabsorption in DCT and collecting duct.

c) Tubular Secretion:

- Adds additional waste to the filtrate from **blood to tubule** (mostly in PCT and DCT).

1. Actively secretes:

- Urea, creatinine, H⁺, K⁺, drugs, toxins**.
- Helps regulate **urine pH** and **electrolyte balance**.
- K⁺ secretion** in **DCT** and **collecting duct** via Na⁺/K⁺ exchange.

3. Counter Current Mechanism:

Counter Current Multiplier (Loop of Henle & Medulla)

- Descending limb**: Permeable to **water**, impermeable to solutes.
- Ascending limb**: Permeable to **Na⁺, K⁺, Cl⁻**, but impermeable to water.
- Ions are **actively transported** out of the ascending limb, making **medulla hypertonic**.

1. Hormones:

- Aldosterone**: Promotes Na⁺ reabsorption.
- ADH**: Promotes water reabsorption.
- Urea**: Further increases medullary osmolarity, aiding water reabsorption.

Counter Current Exchange (Vasa Recta & Interstitial Fluid)

- Vasa recta surrounds loop of Henle.
- Blood flow is **slow**, allowing:
- Water reabsorption** in descending part of vasa recta.
- Solute (Na⁺, Cl⁻) release** in ascending part.
- Maintains osmotic gradient** in medulla, ensuring effective concentration of urine.

Disorders of the Urinary Tract:

1. Urinary Tract Infections (UTIs):

Definition:

Infections affecting any part of the urinary system — **kidneys, ureters, bladder, or urethra** typically caused by **bacteria** (commonly E. coli) or **viruses**.

Common Types of UTIs:

Disorder	Cause	Symptoms
Pyelonephritis (Kidney Infection)	E. coli	Nausea, vomiting, back pain, high fever, chills
Urethritis (Urethra Infection)	E. coli, Chlamydia, Neisseria gonorrhoeae	Burning during urination, painful sex, discharge, frequent urination
Cystitis (Bladder Infection)	E. coli	Urgency, burning urination, cloudy/bloody urine, pelvic pain, low fever





2. Kidney Stones (Renal Calculi / Nephrolithiasis):

Definition:

Hard deposits formed in the kidneys due to accumulation of **salts and minerals** like **calcium, oxalates, uric acid**, etc.

Causes:

- **Hypercalciuria:** high calcium in urine.
- **Hyperoxaluria:** high oxalate in urine.
- **Urinary pH imbalance.**
- **Low fluid intake** or **urinary tract infections.**

Types of Kidney Stones:

Stone Type	Composition	% Incidence	Notes
Calcium Oxalate	Calcium + Oxalate	~80%	Most common
Calcium Phosphate	Calcium + Phosphate	~5–10%	Less common
Uric Acid	Uric acid crystals	~5–10%	Forms in acidic urine
Struvite Stones	Magnesium ammonium phosphate	~10%	Formed in alkaline urine (infection stones)
Cystine Stones	Cystine (amino acid)	<1%	Rare, often in children

Complication:

- **Lithonephritis:** Inflammation of kidney due to irritation from stones.
- **Obstruction** of urine flow.
- **Severe back or abdominal pain.**





Treatment Methods:

Treatment	Description
ESWL (Extracorporeal Shock Wave Lithotripsy)	Non-invasive, uses shock waves to break stones into smaller pieces
PCNL (Percutaneous Nephrolithotomy)	Surgical removal via a small incision using nephroscope and grasper
Laparoscopic Pyelolithotomy	Invasive procedure; kidney is surgically opened to remove large/multiple stones

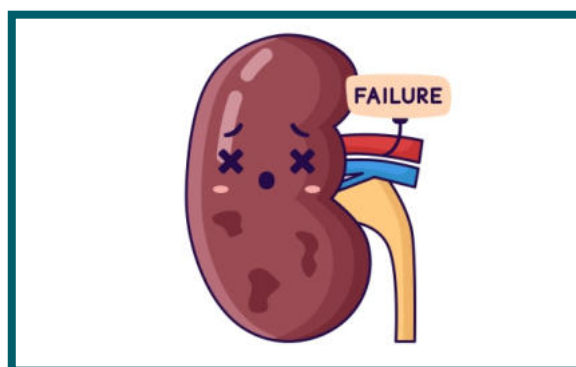
3.Kidney Failure

Definition:

A medical condition where kidneys lose the ability to **filter nitrogenous wastes** and perform **osmoregulatory functions**. It occurs when **>85% of kidney tissue is damaged**.

Types of Kidney Failure:

Type	Description	Causes	Symptoms
Acute Kidney Failure	Sudden loss of kidney function; develops over days.	Blood loss, heart attack, liver failure, severe allergy, dehydration, infection, obstruction	Nausea, decreased urine output, fluid retention, confusion
Chronic Kidney Disease (CKD)	Gradual loss of kidney function over time.	Hypertension, diabetes (hyperglycemia), high cholesterol, kidney cysts, inflammation, stones	Fatigue, short breath, swollen limbs, blood in urine





4. Dialysis

Definition:

Artificial process of **removing waste, toxins, and excess fluids** from blood when kidneys fail to do so.

Types of Dialysis:

1. Hemodialysis (Most common):

- Uses a machine ("artificial kidney") with **semipermeable tubules** in **dialyzing fluid** (no nitrogenous waste).
- **Blood is drawn**, filtered through the machine, and **returned to the body**.
- Requires **AV fistula/graft** (surgically connects artery and vein).
- **Schedule:** 4 hours/session, **3 times/week**.
- **Risks:** Blood infection, thrombosis, internal bleeding, clotting (anticoagulants used).

2. Continuous Renal Replacement Therapy (CRRT):

- Used for **critically ill** patients.
- Runs **24 hours/day** continuously.
- Preferred when the patient cannot tolerate traditional dialysis sessions.

3. Peritoneal Dialysis:

- Uses **abdominal cavity** lined by **peritoneum** as a natural filter.
- **Procedure:**
- Catheter inserted into lower abdomen.
- **Dialysate** enters the cavity ☒ draws toxins from blood ☒ removed after few hours.
- Repeat 4–5 times a day.
- **Advantages:** Flexible, better tolerated, less expensive.
- **Disadvantage:** Risk of **peritonitis** (abdominal infection).



5. Kidney Transplant

Definition:

Surgical procedure to **replace a failed kidney** with a **healthy kidney** from a **living or deceased donor**.

Key Points:

- Transplanted kidney is placed in **lower abdomen** (front side); damaged kidneys often left in place.
- Indicated when **>90% of kidney function is lost**.
- **Causes requiring transplant:**
- Uncontrolled **diabetes (hyperglycemia)**.
- Persistent **hypertension**.
- **Kidney stones**.
- **Polycystic kidney disease**.

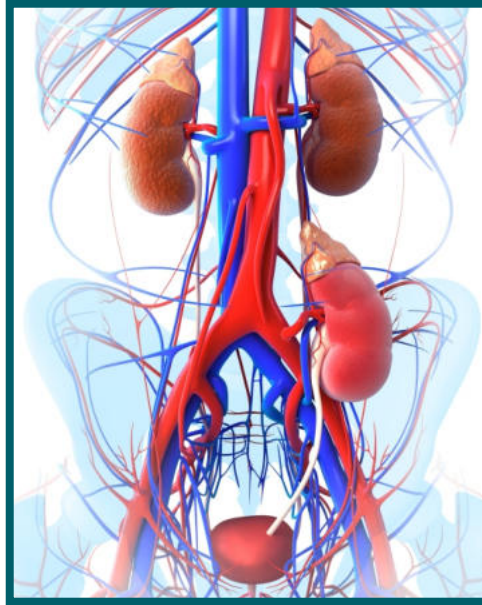




6. Kidney Transplant:

Definition

Kidney transplant is a **major surgical treatment** for **kidney failure** in which a **healthy kidney** from a **living or deceased donor** is transplanted into a patient whose kidneys have lost about **90% of their function**.



Procedure Overview

- A healthy kidney is **implanted in the lower belly (front side)**.
- **Diseased kidneys are usually left in place.**
- **Indications (Causes):**
- **Diabetes (high blood sugar).**
- **Uncontrolled hypertension (high blood pressure).**
- **Polycystic kidney disease.**
- **Kidney stones.**

Principles of Kidney Transplant

- **Blood group compatibility** is essential between donor and recipient.
- **Biochemical compatibility** (including matching of Human Leukocyte Antigen – HLA) must be ensured to reduce the risk of rejection.
- Compatibility minimizes immune response and increases transplant success.

Problems & Complications

- **Immunosuppressant drugs** (e.g., Cyclosporine) are required to:
- Prevent organ rejection.
- **Risk:** Suppression of immune system ☒ Increased vulnerability to **opportunistic infections**.
- **Surgical Complications:**
- **Blood clots.**
- **Bleeding.**
- **Infections** (bacterial or viral).
- **Ureter blockage or leakage.**
- **Transferred diseases or cancer** from donor kidney.
- **Heart attack or stroke.**
- **Death (rare but possible).**





Benefits of Kidney Transplant

- Despite risks, **transplanted patients live longer and have better quality of life** than those on long-term dialysis.
- Allows more **freedom in lifestyle** and **improved health outcomes** compared to dialysis patients.

1. Thermoregulation

- A type of **homeostasis** to maintain a stable **body temperature**.
- **Humans are endotherms:**
- Maintain body temperature around **37°C**, regardless of external conditions.
- Use **behavioral** and **physiological** adaptations to manage heat and cold (e.g., sweating, shivering).

Animals Classification Based on Thermoregulation

2. Ectotherms (Poikilotherms)

- **Definition:** Animals that derive their body temperature from **external environmental sources** (e.g., sunlight, surrounding air).
- **Metabolic Rates:** Ectotherms have **economical** metabolic rates as they do not need to generate internal heat.
- **Behavioral Adaptations:** They rely on **behavioral** means (e.g., basking in the sun, seeking shade) to regulate their body temperature.
- **Habitat:** Commonly found in **constant temperature environments** like the tropics or ocean.

Examples:

- **Invertebrates.**
- **Fishes.**
- **Amphibians.**
- **Reptiles.**



3. Endotherms (Homeotherms)

- **Definition:** Animals that **maintain a constant internal body temperature** independent of the external environment.
- **Metabolic Processes:** They regulate body temperature through **internal metabolic processes**, generating heat internally.
- **Temperature Regulation:** Can maintain stable body temperature even when external conditions are hot or cold through **physiological** (e.g., sweating, shivering) and **behavioral** strategies (e.g., seeking shelter).

Examples:

- **Birds.**
- **Mammals.**





4. Heterothermic Animals

● **Definition:** Animals that can **switch between homeothermic and ectothermic** strategies, adjusting their body temperature based on environmental factors or activity level.

● **Temperature Fluctuations:** They may maintain a constant internal temperature but allow fluctuations when **inactive** (e.g., during sleep or hibernation).

Examples:

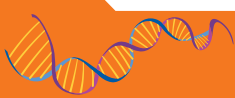
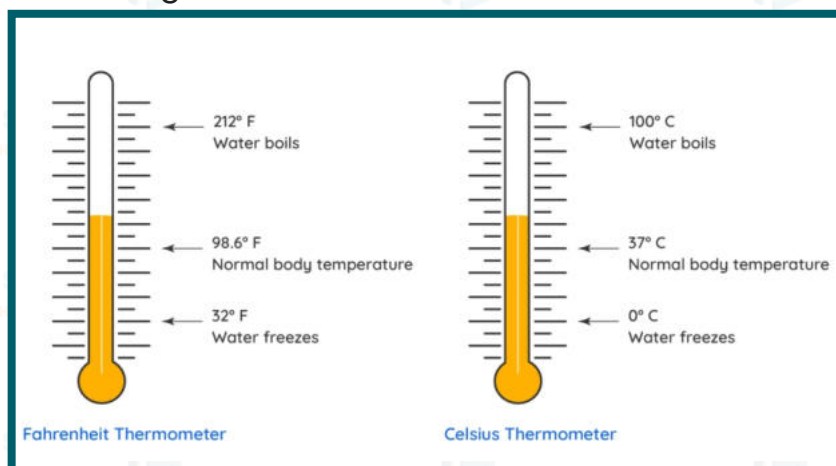
- **Bats** (enter **torpor**—a temporary hibernation-like state)
 - **Hummingbirds** (also enter **torpor** when food is scarce)
 - **Bears** (hibernate during cold months)
- thermoregulation in Humans



1. Optimal Body Temperature

● Humans maintain a **suitable body temperature of 37°C (98.6°F)**.

● **Hypothalamus** in the brain controls thermoregulation, acting as the **thermostat** or **set point** for temperature regulation.





2. Role of Hypothalamus

- The hypothalamus detects changes in body temperature through **receptors** located throughout the body.
- Based on temperature changes, the hypothalamus responds to regulate and maintain the optimal body temperature.

3. Regulation During Hot Conditions (Hyperthermia)

- **Hyperthermia** occurs when body temperature exceeds the optimal 37°C due to external heat or strenuous physical activity.

Process to reduce temperature:

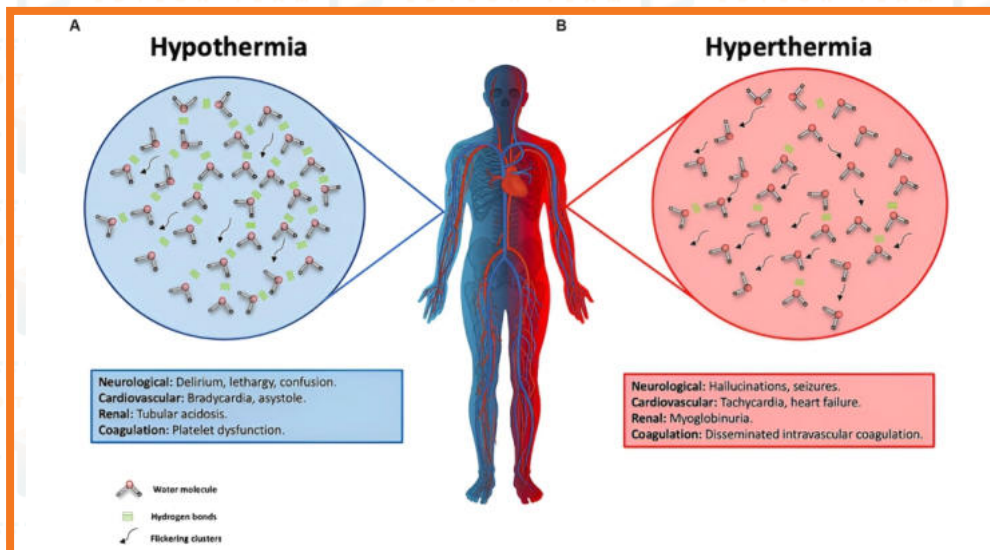
1. Heat produced in the body is transferred to the blood, which then carries it towards the skin.
2. The **hypothalamus** detects this elevated temperature and triggers **sweat glands** to secrete sweat.
3. **Evaporating sweat** absorbs body heat and takes it away from the body into the surrounding environment.
4. **Vasodilation:** Blood vessels near the skin **dilate**, allowing maximum heat transfer away from the body.
5. **Heat transfer:** Conduction (e.g., cold packs) and convection (e.g., moving air or water) also help in heat loss.

4. Regulation During Cold Conditions (Hypothermia)

- **Hypothermia** occurs when the body temperature drops below the normal range.

Process to conserve heat:

1. **Reduced perspiration** and **narrowing of blood vessels** near the skin (called **vasoconstriction**) reduce heat loss.
2. **Thickening of subcutaneous fat** provides insulation to conserve body heat.
3. **Shivering:** Involuntary muscle contractions generate heat to warm the body.



5. Potential Risks

- **Heat Stroke** and **exhaustion** can occur if the body cannot regulate temperature properly in hot conditions (e.g., thinning of hypodermis, excessive heat retention).
- In cold conditions, the body can suffer from **frostbite** or other cold-related injuries if thermoregulation fails.

